

After bursting into existence 13.8 billion years ago in the Big Bang, our Universe was for a time entirely dark because no light-generating objects had yet formed. After a few hundred million years, clumps of matter began to coalesce and heat up, and soon the Universe was bathed in light from the first stars. Today, stars are still the most numerous visible objects in the night sky. We see them as pinpricks

OUT OF THE **DARKNESS**

of light, seeming to differ only in brightness. However, stars are actually extremely diverse, coming in a vast range of sizes and an array of colors. Some will eventually explode to give rise to such strange phenomena as pulsars and black holes. We now also know that many stars, like our Sun, are orbited by planets, some of which might harbor life. Around the time that the first stars ignited, the first galaxies were also forming. Clusters of stars merged into small galaxies, which in turn merged to make bigger galaxies. All the stars visible to the unaided eye are part of our home galaxy, the Milky Way, a structure so vast that light takes a hundred thousand years to cross it. But this galaxy is just one of untold billions of galaxies in the cosmos. Gradually, by using more and more powerful telescopes and other sensing instruments, astronomers are unlocking the secrets of galaxies, along with gaining an understanding of the nature of mysterious phenomena, such as dark matter, in which galaxies seem to be embedded.

◀ Birthplace of stars

The fiery array of stars near the center of this Hubble Space Telescope image is a compact young star cluster some 20,000 light-years away. Called Westerlund 2, it contains some of the hottest, brightest stars known, each with a surface temperature higher than 66,600°F (37,000°C). The cluster lies within a vast, star-forming nebula (cloud of gas and dust) called Gum 29.